Trigonometry

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: The unit circle and radian measure**

Convert 210° to radians.

$210 \times \frac{\pi}{180} = \frac{7\pi}{6}$.

2 **[Foundational]** • **Target: The unit circle and radian measure**

Convert $\frac{5\pi}{4}$ radians to degrees.

$\frac{5\pi}{4} \times \frac{180}{\pi} = 225^\circ$.

3 **[Foundational]** • **Target: The unit circle and radian measure**

Find the exact value of $\sin(240^\circ)$.

240° is in Quadrant III with reference angle 60°, where sine is negative: $\sin(240^\circ) = -\frac{\sqrt{3}}{2}$.

4 **[Foundational]** • **Target: Trigonometric identities**

Simplify $\frac{1-\sin^2\theta}{\cos\theta}$.

$\frac{\cos^2\theta}{\cos\theta} = \cos\theta$.

5 **[Foundational]** • **Target: Trigonometric identities**

Simplify $\tan\theta \cdot \cos\theta$.

$\frac{\sin\theta}{\cos\theta} \cdot \cos\theta = \sin\theta$.

6 **[Foundational]** • **Target: Trigonometric identities**

If $\sin\theta = \frac{3}{5}$ and θ is in Quadrant II, find $\cos\theta$ and $\tan\theta$.

$\cos\theta = -\sqrt{1-\frac{9}{25}} = -\frac{4}{5}$ (negative in Quadrant II). $\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{3/5}{-4/5} = -\frac{3}{4}$.

7 **[Foundational]** • **Target: Double angle formulas**

Given $\cos\theta = \frac{5}{13}$ with θ in Quadrant I, find $\sin 2\theta$ and $\cos 2\theta$.

$\sin\theta = \sqrt{1-\frac{25}{169}} = \frac{12}{13}$. $\sin 2\theta = 2\left(\frac{12}{13}\right)\left(\frac{5}{13}\right) = \frac{120}{169}$. $\cos 2\theta = 1 - 2\sin^2\theta = 1 - \frac{288}{169} = -\frac{119}{169}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Double angle formulas

Use double-angle formulas to express $\sin 4\theta$ in terms of $\sin\theta$ and $\cos\theta$ only.

$\sin 4\theta = 2\sin 2\theta\cos 2\theta = 2(2\sin\theta\cos\theta)(1 - 2\sin^2\theta) = 4\sin\theta\cos\theta(1 - 2\sin^2\theta)$.

9 [Intermediate] • Target: Solving trigonometric equations

Solve $2\sin\theta - 1 = 0$ for $0^\circ \le \theta < 360^\circ$.

$\sin\theta = \frac{1}{2} \Rightarrow \theta = 30^\circ$ or $\theta = 150^\circ$.

10 [Intermediate] • Target: Solving trigonometric equations

Solve $2\cos^2\theta + \cos\theta - 1 = 0$ for $0^\circ \le \theta < 360^\circ$.

Factoring: $(2\cos\theta - 1)(\cos\theta + 1) = 0$. $\cos\theta = \frac{1}{2} \Rightarrow \theta = 60^\circ, 300^\circ$. $\cos\theta = -1 \Rightarrow \theta = 180^\circ$.
Solutions: $60^\circ, 180^\circ, 300^\circ$.

11 [Intermediate] • Target: Graphs of trigonometric functions

The graph shows $y = 3\sin(2x)$. State the amplitude and period.

Amplitude $= 3$. Period $= \frac{2\pi}{2} = \pi$.

12 [Intermediate] • Target: Graphs of trigonometric functions

The function $y = 4\cos(3x - \pi)$ has what amplitude, period, and phase shift?

Amplitude $= 4$. Period $= \frac{2\pi}{3}$. Writing $3x - \pi = 3\left(x - \frac{\pi}{3}\right)$, the phase shift is $\frac{\pi}{3}$ to the right.

13 [Intermediate] • Target: The Law of Sines

In triangle ABC, $a = 10$, $A = 40^\circ$, $B = 65^\circ$. Find b, to the nearest hundredth.

$\frac{b}{\sin B} = \frac{a}{\sin A} \Rightarrow b = \frac{10\sin 65^\circ}{\sin 40^\circ} \approx \frac{9.063}{0.643} \approx 14.10$.

14 [Intermediate] • Target: The Law of Sines

In triangle ABC, $a = 8$, $b = 10$, $A = 35^\circ$. Determine how many triangles are possible, and find angle B (to the nearest tenth of a degree) for each.

$h = b\sin A = 10\sin 35^\circ \approx 5.74$. Since $h < a < b$, two triangles are possible.
$\sin B = \frac{b\sin A}{a} \approx 0.717 \Rightarrow B \approx 45.8^\circ$ or $B \approx 180^\circ - 45.8^\circ = 134.2^\circ$. Both are valid since $A + B < 180^\circ$ in each case.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: The Law of Cosines and the sine area formula • Cross-Lesson: 6.01 Circles & Coordinate Geometry

In triangle ABC shown below, $a = 11$, $b = 14$, and $C = 55^\circ$. Find side c, to the nearest hundredth, and the area of the triangle, to the nearest tenth.

$c^2 = 11^2 + 14^2 - 2(11)(14)\cos 55^\circ \approx 317 - 176.66 = 140.34$, so $c \approx 11.85$. Area $= \frac{1}{2}(11)(14)\sin 55^\circ = 77\sin 55^\circ \approx 63.1$.

16 [Challenging] • Target: The Law of Cosines and the sine area formula • Cross-Lesson: 7.01 Solid Geometry & Similarity

State the two formulas used above (Law of Cosines for a side, and the sine formula for area) in general form for a triangle with sides a, b, c and included angle C.

Law of Cosines: $c^2 = a^2 + b^2 - 2ab\cos C$. Area $= \frac{1}{2}ab\sin C$.

17 [Challenging] • Target: Inverse trigonometric functions • Cross-Lesson: 8.01 Statistics & Probability

Evaluate $\arcsin\left(\frac{\sqrt{2}}{2}\right)$ exactly, in radians.

$\frac{\pi}{4}$.

18 [Challenging] • Target: Inverse trigonometric functions • Cross-Lesson: 1.01 Advanced Algebra

Evaluate $\arccos(-1)$ exactly, in radians.

π.

19 [Challenging] • Target: Inverse trigonometric functions • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

Find $\theta = \arctan(1)$ in degrees (principal value).

45°.

20 [Challenging] • Target: A well-designed word problem: locating a ship from two lighthouses • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

This question has three parts. Two lighthouses A and B are 15 km apart along a straight coastline. A ship at point C is sighted such that $\angle CAB = 72^\circ$ and $\angle CBA = 58^\circ$ (both measured from line AB toward the ship). (a) Find $\angle ACB$. (b) Use the Law of Sines to find the distance AC from lighthouse A to the ship, to the nearest tenth of a km. (c) Find the distance BC from lighthouse B to the ship, to the nearest tenth of a km.

(a) $\angle ACB = 180^\circ - 72^\circ - 58^\circ = 50^\circ$. (b) AC is opposite $\angle B$: $AC = \frac{AB\sin B}{\sin C} = \frac{15\sin 58^\circ}{\sin 50^\circ} \approx 16.6$ km. (c) BC is opposite $\angle A$: $BC = \frac{AB\sin A}{\sin C} = \frac{15\sin 72^\circ}{\sin 50^\circ} \approx 18.6$ km.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: The unit circle and radian measure

Convert 135° to radians.

a. $\frac{3\pi}{5}$

b. $\frac{3\pi}{2}$

c. $\frac{3\pi}{4}$ ✓

d. π

$135 \times \frac{\pi}{180} = \frac{3\pi}{4}$.

22 [Foundational] • Target: Trigonometric identities

Convert 345° to radians.

a. 2π

b. $\frac{23\pi}{13}$

c. $\frac{23\pi}{12}$ ✓

d. $\frac{23\pi}{6}$

$345 \times \frac{\pi}{180} = \frac{23\pi}{12}$.

23 [Foundational] • Target: Double angle formulas

Convert 90° to radians.

a. π

b. $\frac{3\pi}{2}$

c. $\frac{\pi}{3}$

d. $\frac{\pi}{2}$ ✓

$90 \times \frac{\pi}{180} = \frac{\pi}{2}$.

24 [Foundational] • Target: Solving trigonometric equations

Convert 315° to radians.

a. $\frac{7\pi}{5}$

b. $\frac{7\pi}{2}$

c. 2π

d. $\frac{7\pi}{4}$ ✓

$315 \times \frac{\pi}{180} = \frac{7\pi}{4}$.

25 [Foundational] • Target: Graphs of trigonometric functions

Convert 165° to radians.

a. $\frac{11\pi}{6}$

b. $\frac{11\pi}{12}$ ✓

c. π

d. $\frac{11\pi}{13}$

$165 \times \frac{\pi}{180} = \frac{11\pi}{12}$.

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26 [Foundational] • Target: The Law of Sines

Convert 15° to radians.

a. $\frac{\pi}{6}$

b. $\frac{\pi}{13}$

c. $\frac{\pi}{4}$

d. $\frac{\pi}{12}$ ✓

$15 \times \frac{\pi}{180} = \frac{\pi}{12}$.

27 [Foundational] • Target: The Law of Cosines and the sine area formula

Convert 240° to radians.

a. $\frac{5\pi}{3}$

b. $\frac{4\pi}{3}$ ✓

c. π

d. $\frac{8\pi}{3}$

$240 \times \frac{\pi}{180} = \frac{4\pi}{3}$.

28 [Foundational] • Target: Inverse trigonometric functions

Solve $\tan\theta = -\frac{\sqrt{3}}{3}$ for $0^\circ \le \theta < 360^\circ$.

a. $30^\circ, 210^\circ$

b. $150^\circ, 330^\circ$ ✓

c. $135^\circ, 315^\circ$

d. $120^\circ, 300^\circ$

$\tan\theta = -\frac{\sqrt{3}}{3} \Rightarrow \theta = 150^\circ$ or $\theta = 330^\circ$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A well-designed word problem: locating a ship from two lighthouses

Solve $\sin\theta = -\frac{1}{2}$ for $0^\circ \le \theta < 360^\circ$.

a. $225^\circ, 315^\circ$

b. $210^\circ, 330^\circ$ ✓

c. $30^\circ, 150^\circ$

d. $45^\circ, 135^\circ$

$\sin\theta = -\frac{1}{2} \Rightarrow \theta = 210^\circ$ or $\theta = 330^\circ$.

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30 [Intermediate] • Target: Trigonometric Identities & Simplifications

Solve $\cos\theta = \frac{\sqrt{3}}{2}$ for $0^\circ \le \theta < 360^\circ$.

a. $135^\circ, 225^\circ$

b. $30^\circ, 330^\circ$ ✓

c. $120^\circ, 240^\circ$

d. $60^\circ, 300^\circ$

$\cos\theta = \frac{\sqrt{3}}{2} \Rightarrow \theta = 30^\circ$ or $\theta = 330^\circ$.

31 [Intermediate] • Target: Solving Trigonometric Equations

Solve $\sin\theta = \frac{\sqrt{3}}{2}$ for $0^\circ \le \theta < 360^\circ$.

a. $30^\circ, 150^\circ$

b. $240^\circ, 300^\circ$

c. $60^\circ, 120^\circ$ ✓

d. $210^\circ, 330^\circ$

$\sin\theta = \frac{\sqrt{3}}{2} \Rightarrow \theta = 60^\circ$ or $\theta = 120^\circ$.

32 [Intermediate] • Target: Laws of Sines & Cosines with Ambiguous Case

Solve $\tan\theta = \frac{\sqrt{3}}{3}$ for $0^\circ \le \theta < 360^\circ$.

a. $120^\circ, 300^\circ$

b. $30^\circ, 210^\circ$ ✓

c. $135^\circ, 315^\circ$

d. $60^\circ, 240^\circ$

$\tan\theta = \frac{\sqrt{3}}{3} \Rightarrow \theta = 30^\circ$ or $\theta = 210^\circ$.

33 [Intermediate] • Target: Periodic Modeling & Phase Shifts

Solve $\sin\theta = -\frac{\sqrt{3}}{2}$ for $0^\circ \le \theta < 360^\circ$.

a. $210^\circ, 330^\circ$

b. $240^\circ, 300^\circ$ ✓

c. $225^\circ, 315^\circ$

d. $60^\circ, 120^\circ$

$\sin\theta = -\frac{\sqrt{3}}{2} \Rightarrow \theta = 240^\circ$ or $\theta = 300^\circ$.

34 [Intermediate] • Target: The unit circle and radian measure

In triangle ABC, $a = 18$, $A = 80^\circ$, $B = 70^\circ$. Find b, to the nearest hundredth.

a. 17.68

b. 18.18

c. 17.18 ✓

d. 16.68

$\frac{b}{\sin B} = \frac{a}{\sin A} \Rightarrow b = \frac{18\sin 70^\circ}{\sin 80^\circ} \approx \frac{16.914}{0.985} \approx 17.18.$

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35 [Intermediate] • Target: Trigonometric identities

In triangle ABC, $a = 15$, $A = 60^\circ$, $B = 30^\circ$. Find b, to the nearest hundredth.

a. 9.16

b. 8.66 ✓

c. 8.16

d. 9.66

$\frac{b}{\sin B} = \frac{a}{\sin A} \Rightarrow b = \frac{15 \sin 30^\circ}{\sin 60^\circ} \approx \frac{7.500}{0.866} \approx 8.66.$

36 [Intermediate] • Target: Double angle formulas

In triangle ABC, $a = 17$, $A = 80^\circ$, $B = 45^\circ$. Find b, to the nearest hundredth.

a. 12.71

b. 12.21 ✓

c. 13.21

d. 11.71

$\frac{b}{\sin B} = \frac{a}{\sin A} \Rightarrow b = \frac{17 \sin 45^\circ}{\sin 80^\circ} \approx \frac{12.021}{0.985} \approx 12.21.$

37 [Intermediate] • Target: Solving trigonometric equations

In triangle ABC, $a = 13$, $b = 11$, and $C = 130^\circ$. Find side c, to the nearest hundredth, and the area, to the nearest tenth.

a. $21.77, 56.8$

b. $21.77, 54.8$ ✓

c. $20.77, 52.8$

d. $22.77, 54.8$

$c^2 = 13^2 + 11^2 - 2(13)(11)\cos 130^\circ \approx 290 + 183.84 \approx 473.84$, so $c \approx 21.77$. Area $= \frac{1}{2}(13)(11)\sin 130^\circ = \frac{143}{2}\sin 130^\circ \approx 54.8$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Graphs of trigonometric functions • Cross-Lesson: 1.01 Advanced Algebra

In triangle ABC, $a = 9$, $b = 12$, and $C = 70^\circ$. Find side c, to the nearest hundredth, and the area, to the nearest tenth.

a. $12.29, 52.7$

b. $11.29, 48.7$

c. $13.29, 50.7$

d. $12.29, 50.7$ ✓

$c^2 = 9^2 + 12^2 - 2(9)(12)\cos 70^\circ \approx 225 - 73.88 \approx 151.12$, so $c \approx 12.29$. Area $= \frac{1}{2}(9)(12)\sin 70^\circ = 54\sin 70^\circ \approx 50.7$.

39 [Challenging] • Target: The Law of Sines • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

Given $\cos\theta = \frac{24}{25}$ with θ in Quadrant I, find $\sin 2\theta$ and $\cos 2\theta$.

a. $-\frac{336}{625}, \frac{527}{625}$

b. $\frac{336}{625}, \frac{527}{625}$ ✓

c. $\frac{336}{625}, -\frac{527}{625}$

d. $\frac{527}{625}, \frac{336}{625}$

$\sin\theta = \sqrt{1 - \frac{576}{625}} = \frac{7}{25}$. $\sin 2\theta = 2\left(\frac{7}{25}\right)\left(\frac{24}{25}\right) = \frac{336}{625}$. $\cos 2\theta = 1 - 2\sin^2\theta = 1 - \frac{98}{625} = \frac{527}{625}$.

40 [Challenging] • Target: The Law of Cosines and the sine area formula • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

Given $\cos\theta = \frac{12}{13}$ with θ in Quadrant I, find $\sin 2\theta$ and $\cos 2\theta$.

a. $-\frac{120}{169}, \frac{119}{169}$

b. $\frac{120}{169}, -\frac{119}{169}$

c. $\frac{119}{169}, \frac{120}{169}$

d. $\frac{120}{169}, \frac{119}{169}$ ✓

$\sin\theta = \sqrt{1 - \frac{144}{169}} = \frac{5}{13}$. $\sin 2\theta = 2\left(\frac{5}{13}\right)\left(\frac{12}{13}\right) = \frac{120}{169}$. $\cos 2\theta = 1 - 2\sin^2\theta = 1 - \frac{50}{169} = \frac{119}{169}$.

41 [Challenging] • Target: Trigonometric Identities & Simplifications • Cross-Lesson: 4.01 Lines, Triangles & Polygons

Simplify the trigonometric expression: $\frac{1}{1+\sin x} + \frac{1}{1-\sin x}$.

a. $2\sec^2 x$ ✓

b. $2\csc^2 x$

c. $2\cos^2 x$

d. 1

Combine fractions: $\frac{(1-\sin x)+(1+\sin x)}{(1+\sin x)(1-\sin x)} = \frac{2}{1-\sin^2 x} = \frac{2}{\cos^2 x} = 2\sec^2 x$.

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42 [Challenging] • Target: Solving Trigonometric Equations • Cross-Lesson: 6.01 Circles & Coordinate Geometry

Find all solutions to $2\cos^2 x + \cos x - 1 = 0$ in $[0, 2\pi)$.

a. $\frac{\pi}{3}, \frac{5\pi}{3}, \pi$ ✓

b. $\frac{\pi}{6}, \frac{11\pi}{6}, \pi$

c. $\frac{2\pi}{3}, \frac{4\pi}{3}$

d. $\frac{\pi}{4}, \frac{7\pi}{4}$

Factor: $(2\cos x - 1)(\cos x + 1) = 0$. $\cos x = \frac{1}{2} \Rightarrow x = \frac{\pi}{3}, \frac{5\pi}{3}$. $\cos x = -1 \Rightarrow x = \pi$.

43 [Challenging] • Target: Solving Trigonometric Equations • Cross-Lesson: 7.01 Solid Geometry & Similarity

Solve $\tan^2 x = 3$ for $x \in [0, \pi]$.

a. $\frac{\pi}{3}, \frac{2\pi}{3}$ ✓

b. $\frac{\pi}{6}, \frac{5\pi}{6}$

c. $\frac{\pi}{3}$ only

d. $\frac{\pi}{4}, \frac{3\pi}{4}$

$\tan x = \pm\sqrt{3}$. On $[0, \pi]$, $\tan x = \sqrt{3} \Rightarrow x = \frac{\pi}{3}$ and $\tan x = -\sqrt{3} \Rightarrow x = \frac{2\pi}{3}$.

44 [Challenging] • Target: Laws of Sines & Cosines with Ambiguous Case • Cross-Lesson: 8.01 Statistics & Probability

In ΔABC, $a = 5, b = 7,$ and $c = 8$. What is the measure of angle C?

a. 45°

b. 60° ✓

c. 75°

d. 90°

By Law of Cosines: $\cos C = \frac{5^2 + 7^2 - 8^2}{2(5)(7)} = \frac{25 + 49 - 64}{70} = \frac{10}{70} = \frac{1}{7}$ or for 60°: $\frac{25 + 64 - 49}{80} = \frac{40}{80} = \frac{1}{2} \Rightarrow C = 60^\circ$.

45 [Challenging] • Target: Periodic Modeling & Phase Shifts • Cross-Lesson: 1.01 Advanced Algebra

What is the phase shift of the sinusoidal wave $y = 3\sin(2x - \frac{\pi}{3}) + 4$?

a. $\frac{\pi}{6}$ **to the right** ✓

b. $\frac{\pi}{3}$ to the right

c. $\frac{\pi}{6}$ to the left

d. $\frac{\pi}{2}$ to the right

Factor out frequency: $2(x - \frac{\pi}{6})$. The phase shift is $\frac{\pi}{6}$ units to the right.

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